RES POLYTECHNIC, BANGALORE-91

CIE Practice Test II | CSE, Semester III | System and Network Administration (25CS34I) | 50 Marks

Aim: To make a new desktop PC ready for students: identify hardware, test SMPS, record system details, set BIOS boot order, install OS, create a user account and check performance.

Requirements: Desktop PC, screwdriver, anti-static wrist strap, digital multimeter, bootable pen drive (8 GB+), CPU-Z, keyboard, mouse, monitor.

Note: CPU-Z, multimeter and Task Manager values are samples. Write your own system's actual readings in the exam.

a) Hardware identification (10 marks)

Safety: Switch off PC, remove power cord, wear anti-static strap, then open the side panel.

Category	Items and function
Front panel	Power button (ON/OFF), Reset button, Power/HDD LEDs, USB ports (pen drive, mouse), Audio jacks (headphone/mic), Optical drive (CD/DVD)
Rear panel	Power socket (AC mains), PS/2 (old keyboard/mouse), USB, VGA/HDMI/DVI/DisplayPort (monitor), RJ-45 (LAN), Audio jacks (Green: Line-out, Blue: Line-in, Pink: Mic)
Internal components	Motherboard (connects all), CPU + heat sink/fan (processing, cooling), RAM/DIMM (temporary memory), SMPS (AC to DC), HDD/SSD (storage), Expansion cards (graphics/LAN/sound), CMOS battery CR2032 (keeps BIOS settings and clock), BIOS/UEFI chip (firmware, POST)
Storage devices	HDD (SATA 3.5", mechanical, cheap, slow); SSD (SATA 2.5" or M.2 NVMe, fast, no moving parts); Optical drive (CD/DVD)
Cables and connectors	24-pin ATX (motherboard power), 4/8-pin ATX12V (CPU power), SATA data (drive to motherboard), SATA power, Molex 4-pin (old devices), PCIe 6/8-pin (graphics card), front panel connectors (switches, LEDs), front USB/audio headers, power cord, VGA/HDMI and LAN cables

b) SMPS voltage measurement (10 marks, shared with c and d)

Procedure:

1. Switch off PC, remove power cord. Disconnect all SMPS connectors (24-pin, CPU, SATA, Molex).
2. Remove the 4 rear screws and take out the SMPS. **Never open the SMPS cover** (capacitors hold charge).
3. Short **Green (PS_ON)** to any **Black (GND)** wire of the 24-pin connector with a paper clip. Connect mains and switch ON; the fan should spin.
4. Set multimeter to **DC 20 V**. Black probe on GND, red probe on each coloured wire pin; note readings.
5. Switch off, mount the SMPS back, tighten 4 screws, reconnect all connectors, power on and check that POST and boot are normal.

Colour	Voltage	Acceptable range	Measured (sample)
Yellow	+12 V	11.4 to 12.6 V	12.1 V
Red	+5 V	4.75 to 5.25 V	5.04 V
Orange	+3.3 V	3.13 to 3.46 V	3.31 V
Blue	-12 V	-10.8 to -13.2 V	-12.2 V
Purple	+5 V Standby	4.75 to 5.25 V	5.02 V
Grey	PWR_OK (+5 V signal)	About +5 V	5.0 V
Black / Green	GND (0 V) / PS_ON	Green shorted to GND turns SMPS ON	0 V / -

Result: All output voltages are within tolerance; SMPS is working properly.

c) CPU-Z details

Procedure: Install CPU-Z from the CPUID site, open it and note details tab by tab.

Tab	Parameter	Observation (sample)
CPU	Name, code name, package	Intel Core i5-10400, Comet Lake, LGA1200
	Speed / multiplier / bus; cores / threads; voltage	2.9 GHz / x29 / 100 MHz; 6 / 12; 1.0 V
Caches	L1 D / L1 I; L2; L3	6x32 KB / 6x32 KB; 6x256 KB; 12 MB
Mainboard	Manufacturer / model; chipset; BIOS	ASUS / H410M-K; Intel H410; AMI v1405 (2020)
Memory	Type / size / frequency	DDR4 / 8 GB / 2666 MHz

d) BIOS/UEFI and boot order

1. Restart and press **Del / F2 / F10 / F12** (brand dependent) repeatedly as the logo appears.
2. In **Main / System Information** view BIOS version, CPU, RAM size, serial number, date/time.
3. Go to the **Boot** tab, select the pen drive (USB) and move it to **Boot Option #1** using +/- or F5/F6.
4. Press **F10**, then **Yes** (Save and Exit). PC restarts and boots from the pen drive.

Note: UEFI mode needs a GPT/FAT32 pen drive; Legacy mode uses MBR.

Result: System information viewed; pen drive set as first boot device.

e) OS installation: Windows 10/11 (15 marks)

Bootable pen drive: Download Media Creation Tool (or ISO + **Rufus**), insert 8 GB+ pen drive (backup data), choose *Create installation media for another PC* > USB flash drive, wait until ready.

Step	Action
1-2	Insert pen drive, keep USB first in boot order, restart; press any key at "Press any key to boot from USB"
3-4	Select language, time format, keyboard, then **Next, Install now**; enter product key or choose **I don't have a product key**
5-6	Select edition (e.g., Windows 10 Pro), accept license terms, choose **Custom: Install Windows only (advanced)**
7	**Delete** old partitions, select Unallocated space, **New** (set size), **Format**, select partition, **Next**
8-9	Files copy, features install, 2-3 automatic restarts; remove pen drive so it does not boot the installer again
10-11	Setup (OOBE): region, keyboard, network, local account name/password, privacy settings; desktop appears
12	Post-install: drivers, Windows Update, antivirus, check activation

Linux option (Ubuntu): Write ISO with Rufus, boot from USB, *Try or Install Ubuntu*, language/keyboard, Normal installation, *Erase disk and install Ubuntu*, timezone, user/password, Install, restart and remove pen drive.

Result: OS installed successfully from the bootable pen drive.

f) User account CSE_LAB2_01 (10 marks, shared with g)

Command Prompt (Run as Administrator):

```
net user CSE_LAB2_01 Lab@2026#Cse /add
net localgroup Users CSE_LAB2_01 /add
net user CSE_LAB2_01
```

PowerShell (Run as Administrator):

```
$pass = Read-Host -AsSecureString "Enter Password"
New-LocalUser -Name "CSE_LAB2_01" -Password $pass -Description "CSE Lab User"
Add-LocalGroupMember -Group "Users" -Member "CSE_LAB2_01"
```

Strong password: 8-12+ characters with uppercase, lowercase, number and symbol; not like the username. **Verification:** check with `net user` / `Get-LocalUser`, then sign out (Win + L), select CSE_LAB2_01 and log in with the password.

Result: Account created with a strong password and verified by login.

g) Performance check

Task Manager (Ctrl + Shift + Esc): Processes (per-process usage), Performance (live CPU, Memory, Disk, Network, GPU graphs), Startup (boot programs). **Resource Monitor** (Win + R, `resmon`): Overview, CPU, Memory, Disk, Network tabs with process-wise detail.

Resource	Task Manager (sample)	Resource Monitor	Observation
CPU	5-10% idle, 2.9 GHz	Process-wise CPU %, threads	Normal
Memory	3.2 GB / 8 GB (40%)	In use, Available, Reserved	RAM sufficient
Disk	1-3% active time	Read/write per process	No bottleneck
Network	Ethernet 0 Kbps (idle)	Activity per process	Connected

Result: CPU, memory and disk usage are normal; performance is satisfactory.

Tips: Use the format Aim, Requirements, Procedure, Observation, Result. Add screenshots and multimeter photos for the Portfolio (5 marks). Mention safety (power off, anti-static strap) in hardware tasks.